

DESIGN PATTERNS AND PRINCIPLES:

Exercise 1: Implementing the Singleton Pattern

Scenario:

You need to ensure that a logging utility class in your application has only one instance throughout the application lifecycle to ensure consistent logging.

Steps:

1. Create a New Java Project:

- Create a new Java project named **SingletonPatternExample**.

2. Define a Singleton Class:

- Create a class named **Logger** that has a private static instance of itself.
- Ensure the constructor of **Logger** is private.
- Provide a public static method to get the instance of the **Logger** class.

3. Implement the Singleton Pattern:

- Write code to ensure that the **Logger** class follows the Singleton design pattern.

4. Test the Singleton Implementation:

- Create a test class to verify that only one instance of **Logger** is created and used across the application.

CODE:

Logger.java

```
J Logger.java > Logger
1 public class Logger {
2     private static Logger instance;
3
4     private Logger() {
5
6     }
7
8     public static Logger getInstance() {
9         if (instance == null) {
10             instance = new Logger();
11         }
12         return instance;
13     }
14
15     public void log(String message) {
16         System.out.println("Log Message: " + message);
17     }
18 }
```

Main.java

```
J Main.java > Main > main(String[])
1 public class Main {
2     Run | Debug
3     public static void main(String[] args) {
4         Logger logger1 = Logger.getInstance();
5         Logger logger2 = Logger.getInstance();
6
7         logger1.log(message: "This is the first log message.");
8         logger2.log(message: "This is the second log message.");
9
10        if (logger1 == logger2) {
11            System.out.println(x: "Both logger1 and logger2 are the same instance.");
12        }
13        else {
14            System.out.println(x: "logger1 and logger2 are different instances.");
15        }
16    }
17 }
```

OUTPUT:

```
● PS C:\Users\sunke\Desktop\Exercise 1 Implementing the Singleton Pattern> javac Main.java
● PS C:\Users\sunke\Desktop\Exercise 1 Implementing the Singleton Pattern> java Main
Log Message: This is the first log message.
Log Message: This is the second log message.
Both logger1 and logger2 are the same instance.
```